

MODULE 1

Security Attacks - Security Services; Security Mechanisms; Fundamental Security Design Principles; Cryptography - Symmetric Cipher Model, Substitution Techniques, Transposition techniques; Traditional Block Cipher Structure.

1.1 Computer Security Concepts

Computer Security: The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability, and confidentiality of information system resources (includes hardware, software, firmware, information/data, and telecommunications).

This definition introduces three key objectives that are at the heart of computer security:

- **Confidentiality:** This term covers two related concepts:
 - Data confidentiality: Assures that private or confidential information is not made available or disclosed to unauthorized individuals.
 - Privacy: Assures that individuals control or influence what information related to them may be collected and stored and by whom and to whom that information may be disclosed.
- **Integrity:** This term covers two related concepts:
 - Data integrity: Assures that information and programs are changed only in a specified and authorized manner.
 - System integrity: Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.
- **Availability:** Assures that systems work promptly and service is not denied to authorized users.

1.1.1 The challenges of computer security

1. The major requirements for security services can be given self-explanatory, one-word labels: confidentiality, authentication, nonrepudiation, or integrity.

2. In developing a particular security mechanism or algorithm, one must always consider potential attacks on those security features.
3. When the various aspects of the threat are considered that elaborate security mechanisms make sense.
4. It is necessary to decide where to use the security mechanisms, both in terms of physical placement(at what points in a network) and in a logical sense(at what layer or layers of a architecture such as TCP/IP)
5. Security mechanism typically involve more than a particular algorithm or protocol.
6. Computer and network security is essentially a battle of wits between a attackers and the designer/ administrator.
7. Security requires regular, even constant, monitoring.
8. Security is still too often an afterthought to be incorporated into a system after the design is complete rather than being an integral part of the design process.
9. Many users and even security administrator view strong security as an impediment to efficient and user-friendly operation of an information system.

1.2 THE OSI SECURITY ARCHITECTURE

The OSI security architecture focuses on security attack, security mechanism , security service ;

security attack: Any action that compromises the security of information owned by an organization.

Security mechanism: A process that is designed to detect, prevent, or recover from a security attack.

Security service: A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization.

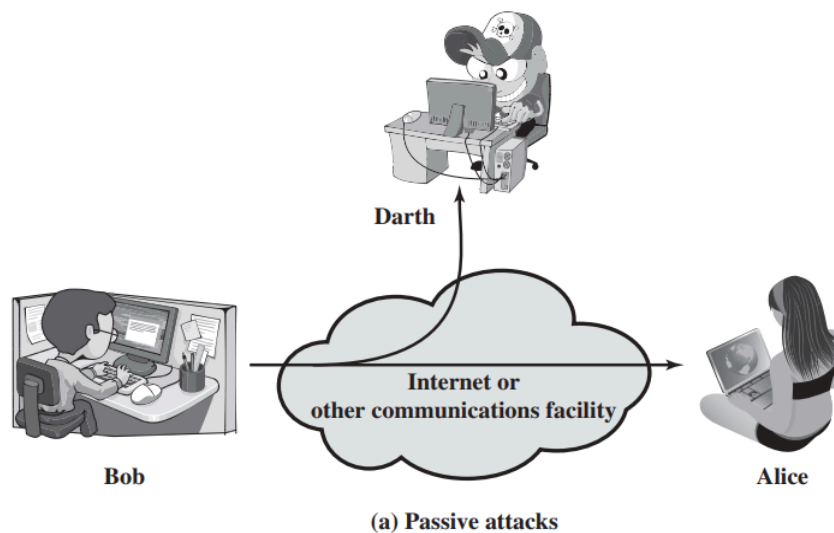
1.3 SECURITY ATTACKS

A useful means of classifying security attacks, used both in X.800 and RFC 2828, is in terms of *passive attacks* and *active attacks*.

1.3.1 Passive Attack

A **passive attack** attempts to learn or make use of information from the system but does not affect system resources.

A **passive attack** is an attack in which an attacker **only monitors or listens to communication** without modifying the data. The goal is to obtain information secretly.



Two types of passive attacks are:

1. **release of message contents** - The attacker gains access to the actual content of transmitted data.
 - Sensitive information such as:
 - Telephone conversations
 - E-mail messages
 - File transfers
 - Confidential documents
 - The objective is to prevent unauthorized users from reading these messages.

Example:

- An attacker intercepts an email containing bank account details.

2. **traffic analysis** - Even if the message is encrypted, attackers can still analyze communication patterns.
 - They cannot read the message, but they can observe:
 - a. Sender and receiver identities
 - b. Location of communicating devices
 - c. Frequency of communication
 - d. Message size (length)
 - e. Time of communication
 - This information may help attackers guess the nature of the communication.

Example:

- During military operations, frequent encrypted communication between two bases may indicate an upcoming mission, even though the message contents remain hidden.

Passive attacks are **very difficult to detect** because:

- Data is not altered.
- Network operation appears normal.
- Neither sender nor receiver notices the attack.

1.3.2 Active Attack

Active attacks involve some modification of the data stream or the creation of a false stream and can be subdivided into four categories: *masquerade*, *replay*, *modification of messages*, and *denial of service*.

1. A **masquerade** takes place when one entity pretends to be a different entity (path 2 of Figure b is active). A masquerade attack usually includes one of the other forms of active attack.

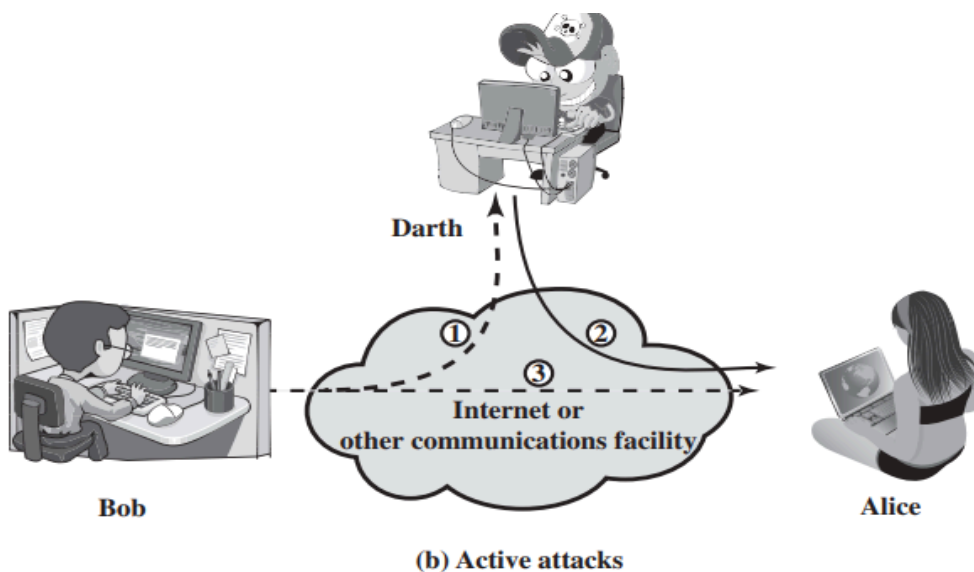
For example, authentication sequences can be captured and replayed after a valid authentication sequence has taken place, thus enabling an authorized entity

with few privileges to obtain extra privileges by impersonating an entity that has those privileges.

2. **Replay** involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect (paths 1, 2, and 3 active).
3. **Modification of messages** simply means that some portion of a legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect (paths 1 and 2 active).

For example, a message meaning “Allow John Smith to read confidential file accounts” is modified to mean “Allow Fred Brown to read confidential file accounts.”

4. The **denial of service** prevents or inhibits the normal use or management of communications facilities (path 3 active). This attack may have a specific target; for example, an entity may suppress all messages directed to a particular destination (e.g., the security audit service). Another form of service denial is the disruption of an entire network, either by disabling the network or by overloading it with messages so as to degrade performance.



1.4 Security Mechanisms

Security mechanisms are **techniques and methods used to implement security services** such as confidentiality, integrity, authentication, and access control in computer networks.

According to the **X.800 Security Architecture**, security mechanisms are classified into:

1. **Specific Security Mechanisms:** These mechanisms are implemented within a particular protocol layer (such as TCP/IP or the application layer) to provide specific security services.
2. **Pervasive Security Mechanisms:** These mechanisms are not tied to any specific protocol layer and support overall system security.

Specific Security Mechanisms	Pervasive Security Mechanisms
Encipherment The use of mathematical algorithms to transform data into a form that is not readily intelligible. The transformation and subsequent recovery of the data depend on an algorithm and zero or more encryption keys.	Trusted Functionality That which is perceived to be correct with respect to some criteria (e.g., as established by a security policy).
Digital Signature Data appended to, or a cryptographic transformation of, a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery (e.g., by the recipient).	Security Label The marking bound to a resource (which may be a data unit) that names or designates the security attributes of that resource.
Access Control A variety of mechanisms that enforce access rights to resources.	Event Detection Detection of security-relevant events

Data Integrity A variety of mechanisms used to assure the integrity of a data unit or stream of data units.	Security Audit Trail Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.
Authentication Exchange A mechanism intended to ensure the identity of an entity by means of information exchange.	Security Recovery Deals with requests from mechanisms, such as event handling and management functions, and takes recovery actions.
Traffic Padding The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.	
Routing Control Enables selection of particular physically secure routes for certain data and allows routing changes, especially when a breach of security is suspected.	
Notarization The use of a trusted third party to assure certain properties of a data exchange.	

1.5 Fundamental Security Design Principles

1.6 Cryptography - Symmetric Cipher Model, Substitution Techniques, Transposition techniques

Cryptography is the science of protecting information by converting readable data into an unreadable form so that only authorized users can access it.

Important Terminologies

1. Plaintext: Plaintext is the **original readable message or data** before encryption.

Example: Meet me at 10 AM

2. Ciphertext: Ciphertext is the **encrypted (coded) form** of the plaintext. It cannot be understood without decryption.

Example: X9K#P2@LMQ

3. Encryption (Enciphering): Encryption is the process of converting **plaintext into ciphertext** using an encryption algorithm and a key. It protects data from unauthorized access. Also it ensures confidentiality.

Flow

Plaintext → Encryption → Ciphertext

4. Decryption (Deciphering): **Decryption** is the process of converting **ciphertext back into plaintext** using the correct decryption key. It allows authorized users to read the original message.

Flow

Ciphertext → Decryption → Plaintext

5. Cryptography: Cryptography is the study and practice of securing information through encryption and decryption techniques.

Purpose

- Maintain confidentiality.
- Ensure secure communication.
- Protect sensitive information.

Example: AES, RSA, DES

6. Cryptographic System (Cipher): A cryptographic system or cipher is the algorithm or method used for encryption and decryption.

Example: Caesar Cipher, AES Cipher, RSA Cipher

7. Cryptanalysis: Cryptanalysis is the process of **breaking encrypted messages without knowing the encryption key**. It is commonly known as **code breaking**.

Purpose

- Recover the original message.
- Analyze weaknesses in encryption algorithms.

Example: An attacker attempts to decrypt an encrypted message without the secret key.

8. Cryptology: **Cryptology** is the combined study of: Cryptography and Cryptanalysis

1.6.1 Symmetric Cipher Model

A symmetric encryption scheme has five ingredients

1. **Plaintext:** This is the original intelligible message or data that is fed into the algorithm as input.
2. **Encryption algorithm:** The encryption algorithm performs various substitutions and transformations on the plaintext.
3. **Secret key:** The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.
4. **Ciphertext:** This is the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will

produce two different ciphertexts. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.

5. **Decryption algorithm:** This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

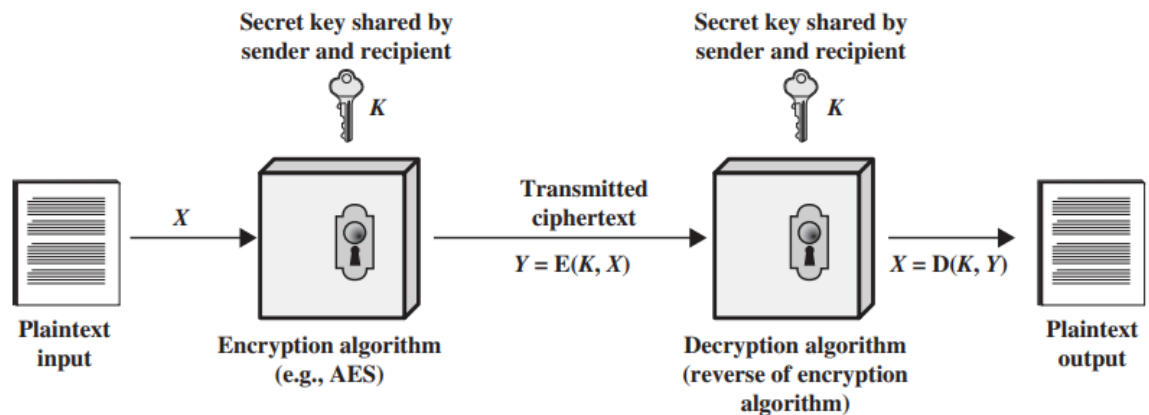


Figure 2.1 Simplified Model of Symmetric Encryption

Requirements for Secure Use of Conventional Encryption

1. Strong Encryption Algorithm

- The encryption algorithm should be difficult to break.
- Even if an attacker knows the algorithm and has access to ciphertext (or plaintext–ciphertext pairs), they should not be able to find the plaintext or the secret key.

2. Secure Secret Key

- The sender and receiver must securely share the secret key.
- The key must remain confidential.
- If the secret key is exposed, all encrypted messages can be read.

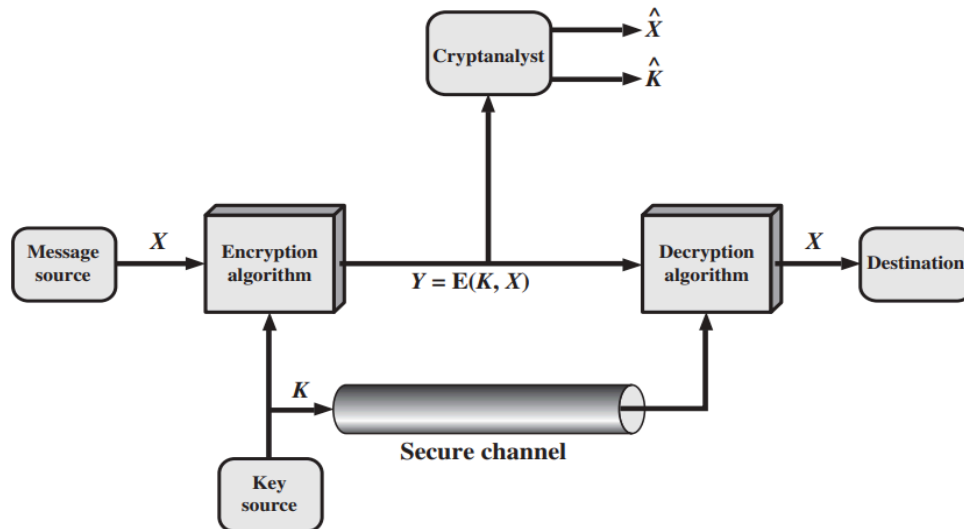


Figure 2.2 Model of Symmetric Cryptosystem

In the **conventional encryption model**,

- The sender starts with a **plaintext message (X)** and encrypts it using a **secret key (K)** and an **encryption algorithm (E)** to produce **ciphertext (Y)**.
- The ciphertext is transmitted over the communication channel.
- At the receiver's end, the same secret key is used with the **decryption algorithm (D)** to convert the ciphertext back into the original plaintext.
- The encryption process is represented as $Y = E(K, X)$, and the decryption process as $X = D(K, Y)$.
- The security of the system depends on keeping the **secret key confidential**, even though the encryption and decryption algorithms are publicly known.
- An attacker may try to recover either the plaintext or the secret key from the intercepted ciphertext.

Cryptography

Cryptographic systems are characterized along three independent dimensions:

- First, they are classified by the **type of operations** used, namely **substitution**, where each plaintext element is replaced with another element, and **transposition**, where the

positions of the plaintext elements are rearranged. Most modern encryption algorithms combine both techniques and are called **product ciphers**.

- Second, cryptographic systems are classified by the **number of keys used**. **Symmetric encryption** uses the same key for both encryption and decryption, whereas **asymmetric encryption** uses two different keys—a public key and a private key.
- Third, they are classified by the **way the plaintext is processed**. A **block cipher** encrypts data in fixed-size blocks, while a **stream cipher** encrypts data continuously, one bit or byte at a time.

There are two general approaches to attacking a conventional encryption scheme:

- **Cryptanalysis:** Cryptanalytic attacks rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext–ciphertext pairs. This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.
- **Brute-force attack:** The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.

Table 2.1 Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

1.6.2 Substitution Techniques

A **substitution technique** is a classical encryption method in which each letter (or group of letters) of the plaintext is **replaced by another letter, number, or symbol** according to a substitution rule. The order of letters remains the same; only their identities change.

Characteristics:

- Replaces plaintext symbols with ciphertext symbols.
- The position of letters remains unchanged.
- Encryption and decryption use predefined substitution rules.

Types of Substitution Techniques

1. Caesar Cipher (Shift Cipher): The **Caesar Cipher** is the simplest substitution cipher. Each letter of the plaintext is shifted by a fixed number of positions in the alphabet.

Encryption Formula: $C = (P + k) \bmod 26$ Where

- **P** = Plaintext letter
- **C** = Ciphertext letter
- **k** = Shift value

Decryption Formula: $P = (C - k) \bmod 26$

Encryption Algorithm

1. Choose a shift value **k**.
2. Convert each plaintext letter into its numerical equivalent (A=0, B=1, ..., Z=25).
3. Add **k** to each letter.
4. Take modulo 26.
5. Convert numbers back to letters.

Decryption Algorithm

1. Convert ciphertext letters to numbers.
2. Subtract **k**.
3. Take modulo 26.
4. Convert back to plaintext.

Example:

Plaintext: HELLO Shift = 3 Ciphertext

Plaintext H E L L O

Shift +3 +3 +3 +3 +3

Ciphertext K H O O R

Advantages

- Very simple.
- Easy to implement.

Disadvantages

- Only 25 possible keys.
- Easily broken using brute-force attack.

2. Monoalphabetic Cipher: A **Monoalphabetic Cipher** uses a **fixed substitution alphabet**. Every occurrence of a plaintext letter is replaced by the same ciphertext letter.

Encryption Algorithm

1. Create a substitution table.
2. Replace every plaintext letter using the table.
3. Produce ciphertext.

Decryption Algorithm

1. Reverse the substitution table.
2. Replace ciphertext letters with their original letters.

Plain	A	B	C	D	E
--------------	----------	----------	----------	----------	----------

Cipher	Q	W	E	R	T
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Plain text: BAD

Cipher text:WQR

Advantages

- More secure than Caesar cipher.
- Large number of possible keys.

Disadvantages

- Vulnerable to **frequency analysis** because one plaintext letter always maps to the same ciphertext letter.

3. Playfair Cipher: The **Playfair Cipher** encrypts **pairs of letters (digraphs)** instead of single letters, making it more secure than simple substitution ciphers.

Key Matrix Construction

1. Choose a keyword.
2. Remove repeated letters.
3. Fill a **5 × 5 matrix** with the keyword.
4. Fill remaining letters of the alphabet.
5. Combine **I** and **J** into one cell.

Example keyword:

MONARCHY

Matrix

M O N A R

C H Y B D

E F G I K

L P Q S T

U V W X Z

Encryption Rules

- **Rule 1:** If both letters are in the **same row**, replace each with the letter to its right.
- **Rule 2:** If both letters are in the **same column**, replace each with the letter below it.
- **Rule 3:** Otherwise, form a rectangle and replace each letter with the letter in the same row but in the other column.

Encryption Algorithm

1. Construct the key matrix.
2. Divide plaintext into pairs.
3. Insert **X** if identical letters appear together.
4. Apply Playfair rules.
5. Produce ciphertext.

Example: Plaintext - HI

Ciphertext depends on the key matrix.

Advantages

- Encrypts letter pairs.
- More secure than monoalphabetic substitution.

Disadvantages

- Frequency analysis is still possible.
- Matrix construction is time-consuming.

4. Hill Cipher: The **Hill Cipher** is a polygraphic substitution cipher that encrypts blocks of letters using **matrix multiplication** and **modulo 26 arithmetic**.

Encryption Formula : $C = KP \bmod 26$ Where

- **K** = Key matrix
- **P** = Plaintext vector
- **C** = Ciphertext vector

Decryption Formula : $P = K^{-1} C \bmod 26$

Encryption Algorithm

1. Choose an invertible key matrix.
2. Convert plaintext into numerical vectors.
3. Multiply the key matrix by the plaintext vector.
4. Take modulo 26.
5. Convert numbers into letters.

Example

Plaintext: ACT

Key Matrix

[6 24 1]

| 13 16 10 |

[20 17 15]

Ciphertext: POH

Advantages

- Stronger than simple substitution ciphers.
- Encrypts multiple letters simultaneously.

Disadvantages

- Requires knowledge of matrices.
- Key matrix must have an inverse modulo 26.

5. Polyalphabetic Cipher (Vigenère Cipher): A Vigenère Cipher uses multiple substitution alphabets controlled by a keyword. Each plaintext letter may be encrypted differently depending on the corresponding key letter.

Encryption Formula: $C=(P+K)\bmod 26$

Decryption Formula: $P=(C-K)\bmod 26$

Encryption Algorithm

1. Choose a keyword.
2. Repeat the keyword until it matches the plaintext length.
3. Convert plaintext and key letters to numbers.
4. Add corresponding values modulo 26.
5. Convert back to letters.

Example

Plaintext: ATTACKATDAWN

Key :LEMONLEMONLE

Ciphertext: LXFOPVEFRNHR

Advantages

- More secure than monoalphabetic substitution.
- Reduces the effectiveness of frequency analysis.

Disadvantages

- If the key is short or repeated, it can be broken using cryptanalysis (e.g., Kasiski examination)

1.6.3 Transposition techniques; A **transposition cipher** is an encryption technique in which the **positions of the plaintext letters are changed**, while the letters themselves remain unchanged.

Characteristics

- Rearranges the order of letters.
- Does not replace letters.
- Encryption and decryption depend on the arrangement pattern.

1. Rail Fence Cipher

The **Rail Fence Cipher** is the simplest transposition cipher. The plaintext is written diagonally across a fixed number of rows (rails), and the ciphertext is obtained by reading the rows one by one.

Encryption Algorithm

1. Choose the number of **rails (rows)**.
2. Write the plaintext diagonally in a zigzag pattern across the rails.
3. After reaching the last rail, continue diagonally upward.
4. Read the characters row by row to obtain the ciphertext.

Decryption Algorithm

1. Determine the number of rails used.
2. Mark the zigzag pattern for the ciphertext.
3. Fill the ciphertext letters into the pattern row by row.
4. Read the letters diagonally to recover the plaintext.

Example

Plaintext: MEETMEAFTERTHETOGAPARTY

Step 1: Write in Zigzag Pattern (2 Rails)

M E M A T R H T G P R Y

E T E F E T E O A A T

Step 2: Read Row by Row

Ciphertext: MEMATRHTGPRYETEFETEOAAT

Advantages

- Very easy to understand and implement.
- Requires only rearrangement of letters.

Disadvantages

- Provides very weak security.
- Easily broken by trying different numbers of rails.

2. Row Transposition Cipher:

In a **Row Transposition Cipher**, the plaintext is written row by row into a table with a fixed number of columns. The columns are then read according to a **key (permutation)** to produce the ciphertext.

EncryptionAlgorithm

1. Choose a numeric key.
2. Write the plaintext row by row in a table.
3. Number the columns according to the key.
4. Read the columns in the order specified by the key.
5. Combine the columns to obtain the ciphertext.

Decryption Algorithm

1. Determine the table size.
2. Fill the ciphertext column by column according to the key.
3. Read the table row by row.
4. Recover the plaintext.

Example

Key: 4 3 1 2 5 6 7

Plaintext : ATTACKPOSTPONEDUNTILTWOAM

Arrange the plaintext in rows:

4 3 1 2 5 6 7

A T T A C K P

O S T P O N E

D U N T I L T

W O A M X X X

Read the columns in the order of the key (1, 2, 3, 4, 5, 6, 7) to obtain the ciphertext.

Advantages

- More secure than the Rail Fence cipher.

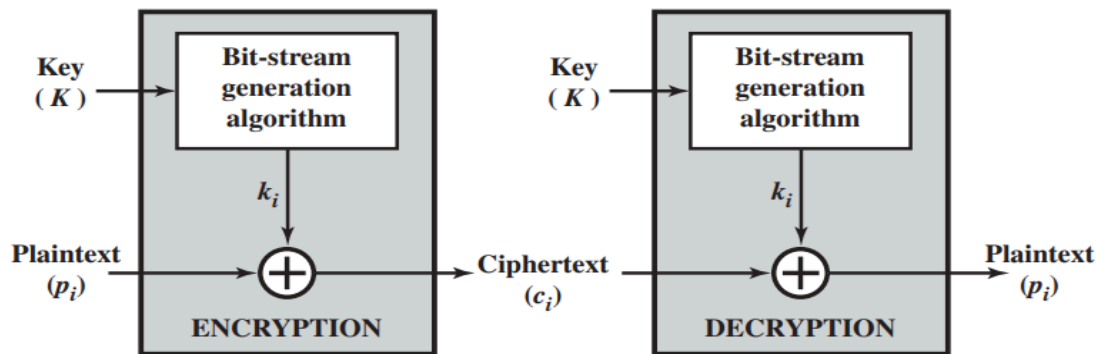
- Harder to break because of column permutations.

Disadvantages

- Can be broken using statistical analysis if the key is short.
- Less secure than modern encryption algorithms

1.7 Traditional Block Cipher Structure

1.Stream Ciphers: A **stream cipher** is a symmetric encryption technique that encrypts a digital data stream **one bit or one byte at a time** using a continuously generated **keystream**.



(a) Stream cipher using algorithmic bit-stream generator

Working Principle

- A **keystream** is generated using a secret key.
- Each plaintext bit is combined with the corresponding keystream bit using the **XOR (Exclusive-OR)** operation.
- The result is the ciphertext.

Encryption Formula: $C_i = P_i \oplus K_i$ Where:

- P_i = Plaintext bit

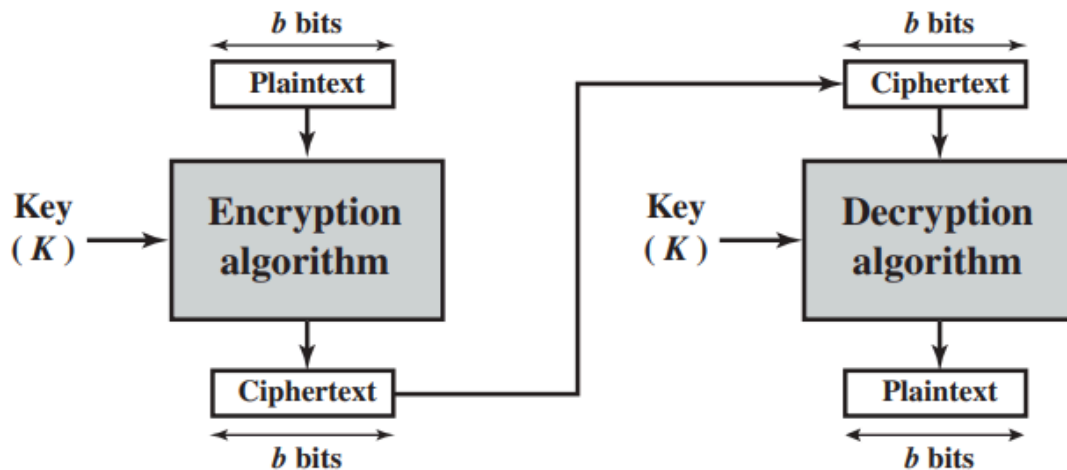
- K_i = Keystream bit
- C_i = Ciphertext bit
- $\oplus$ = XOR operation

Decryption Formula: $P_i = C_i \oplus K_i$

2. Block Ciphers : A block cipher encrypts a **fixed-size block of plaintext** as a whole and produces a **ciphertext block of equal length**.

Working Principle

- Plaintext is divided into **fixed-size blocks** (e.g., 64 or 128 bits).
- Each block is encrypted using a **shared secret key**.
- The receiver decrypts each ciphertext block using the **same key**



(b) Block cipher

Characteristics

- Encrypts **fixed-size blocks** of data.

- Uses a **single secret key** (symmetric encryption).
- Plaintext and ciphertext blocks have the **same length**.
- Common block sizes are **64 bits** and **128 bits**.
- Can also operate like a **stream cipher** using different modes of operation.

Motivation for the Feistel Cipher Structure

Definition

The **Feistel Cipher Structure** was developed to create a **secure and efficient block cipher** that allows the **same algorithm** to be used for both encryption and decryption.

Why Feistel Structure?

- Simplifies the design of secure block ciphers.
- Uses repeated rounds to improve security.
- Encryption and decryption use the same structure.
- Only the order of subkeys is reversed during decryption.
- Supports substitution and permutation to provide **confusion** and **diffusion**.

Design Goals

- Easy implementation.
- Strong resistance against cryptanalysis.
- Efficient hardware and software implementation.

The Feistel Cipher

A **Feistel Cipher** is a symmetric block cipher structure in which the plaintext block is divided into **two equal halves**, and multiple rounds of processing are performed using different round keys.

Feistel Structure

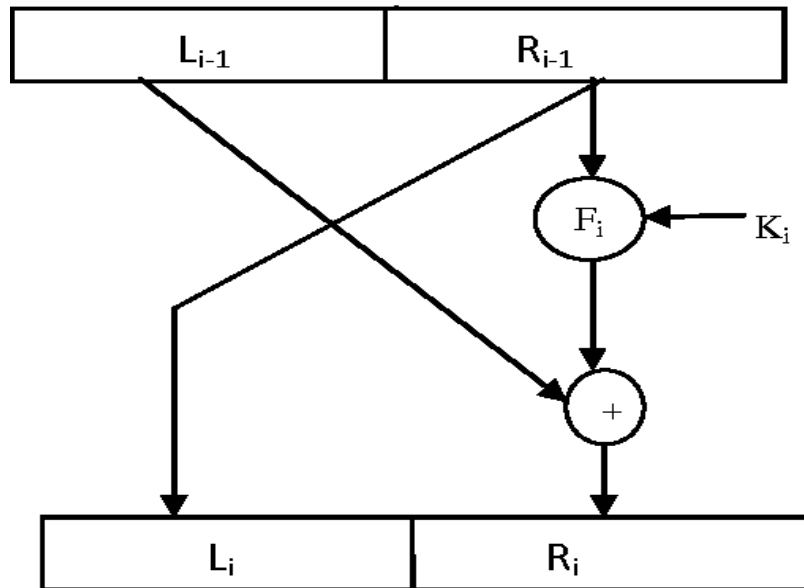


Fig 2: Single Round in Feistel cipher Structure

Feistel Round

For each round:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Where:

- L_i = Left half after round i
- R_i = Right half after round i
- K_i = Round key
- F = Round function

Encryption Algorithm

1. Divide the plaintext block into left and right halves.

2. Generate the required round keys.
3. Apply the Feistel round repeatedly.
4. Swap the two halves after each round.
5. After the final round, combine the halves to produce the ciphertext.

Decryption Algorithm

1. Divide the ciphertext into left and right halves.
2. Use the **same Feistel structure**.
3. Apply the round keys in **reverse order**.
4. Combine the final halves to recover the plaintext.

Advantages of Feistel Cipher

- Same structure for encryption and decryption.
- Easy to implement.
- Highly secure with multiple rounds.
- Basis of many modern block ciphers.